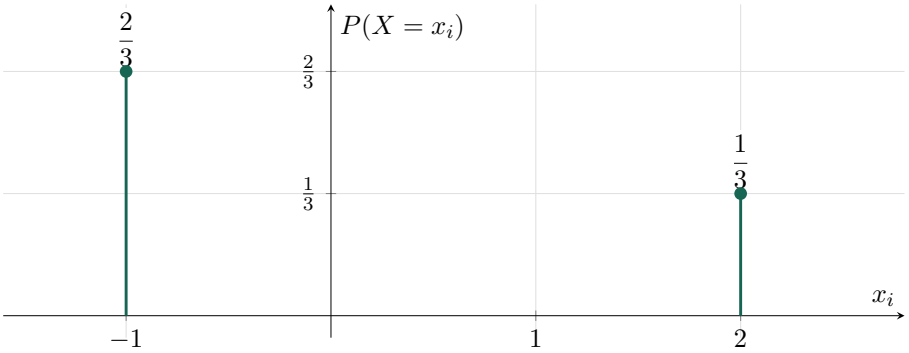
8.1 Variable, event and law

Definition and results

On a finite universe, a real random variable X is a function that combines a number real at every exit. $\{X = a\}$ collects the issues giving value a. Its law with each separate value x_i probability $p_i = P(X = x_i)$, with $p_i \geq 0$ and $\sum p_i = 1$.

Status and justification. Definitions. Events associated with separate values form a partition of the universe; this justifies the sum of probabilities.

Example solved. A balanced dice gives an X gain equal to 2 for 1 or 2, and to -1 If not. The law is $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$. $\{X \leq 0\} = \{3; 4; 5; 6\}$: Probability $2\frac{-}{3}$.



The height indicates the probability: $P(X = -1) = \frac{2}{3}$ and $P(X = 2) = \frac{1}{3}$. Accurate probabilities are on both sticks.

Point of vigilance

Values are not always equiprobable even if the outcomes are. which give the same gain.

Immediate application. Two balanced pieces are launched. X counts batteries. Write his law and $P(X \leq 1)$.

8.2 Expectation and linearity

Definition and results

Expectation is $E(X) = \sum_i x_i p_i$. It describes the theoretical long-term average, not a result
For variables X and Y on the same finite universe
and real a,b: $E(aX + b) = aE(X) + b$ and $E(X + Y) = E(X) + E(Y)$, no hypothesis
independence.

Status and justification. Sleeping on the exits ω, $E(X + Y) = \sum_\omega (X(\omega) + Y(\omega))P(\omega)$; distribute the sum gives the result. The constant b has expectation b because the probabilities have sum 1.

Example solved. For the previous law $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$, $E(X) = 0$: the game is fair if X is the net gain. $Y = 3X - 2$, $E(Y) = 3 \times 0 - 2 = -2$.

Point of vigilance

A expectation is not an individual guarantee; a game can be lost on every horizon
in spite of a positive expectation.

Immediate application. Gross gain equals 10 with probability $\frac{1}{5}$ and 0 otherwise. What bet makes the fair play?

8.3 Variance, standard deviation and König·Huygens

Definition and results

Noting $\mu = E(X)$, $V(X) = E((X - \mu)^2) = \sum_i (x_i - \mu)^2 p_i$. The standard deviation is $\sigma(X) = \sqrt{V(X)}$.
The formula of König·Huygens gives $V(X) = E(X^2) - E(X)^2$. A variance is always
positive or zero; standard deviation has the same unit as X, variance a unit squared.

Status and justification. Develop ($X - \mu)^2 = X^2 - 2\mu X + \mu^2$, apply linearity and $E(X) = \mu$: $E(X^2) - 2\mu^2 + \mu^2$. We get the formula.

Example solved. For X values 0 and 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$, $E(X) = 1$, $E(X^2) = 4$, $V(X) = 3$ and $\sigma = \sqrt{3}$. Direct calculation: $(0 - 1)^2(\frac{3}{4}) + (4 - 1)^2(\frac{1}{4}) = 3$.

Point of vigilance

$E(X^2)$ is not $E(X)^2$. A calculated negative variance indicates an error or a very small one
error to be dealt with explicitly.

Immediate application. For X values -1, 1 equiprobable, calculate expectation, variance and Standard deviation.

8.4 Transform and Compare Gains

Definition and results

For real a,b, $V(aX+b) = a^2V(X)$ and $\sigma(aX+b) = |a|\sigma(X)$ A translation moves the average without changing the dispersion; a multiplication changes the scale. Two sets of the same expectation can have very different dispersions.

Status and justification. $aX+b-E(aX+b) = a(X-E(X))$. Raising squared and then taking expectation, we get $a^2V(X)$ The square root of a^2 is $|a|$.

Example solved. A game A always returns 2. A game B earns 0 or 4 equibably.
Two have expectation 2; variances 0 and 4. Their distributions are therefore not interchangeable despite the same average.

Point of vigilance

The variance of a sum is not, in general, the sum of the variances; no rule of this type n is used without assumptions.

Immediate application. $E(X) = 3$, $V(X)$ = 4. Give the characteristics of $Y = -2X + 5$.

8.5 Samples and mean observed

Definition and results

A sample size n is here a list of independent achievements of the same variable X. The mean observed m varies from sample to sample. $\mu = E(X)$; of repeats make it possible to observe the proportion of satisfactory samples $|m-\mu| \leq \frac{2\sigma}{\sqrt{n}}$. This threshold is an experimental task of the programme, not an assertion of coverage rates Right.

Status and justification. Experimentation: here no convergence theorem or law is shown
The list size, the calculation of the average and the distinction between n and the number are checked. No samples.

Example solved. For X of 0 and 2 equiprobable values, $\mu = 1$, σ = 1. For n=100, threshold $\frac{2}{10} = 0{,}2$: averages are counted in [0 ,8; 1,A result of 956 out of 1000 means frequency 0.956 for this simulation only.

Point of vigilance

Do not copy a single draw once; do not announce that the proportion is necessarily valid 95 %.

Immediate application. For $\mu = 5$, σ = 3 and n=36, give the experimental interval requested.